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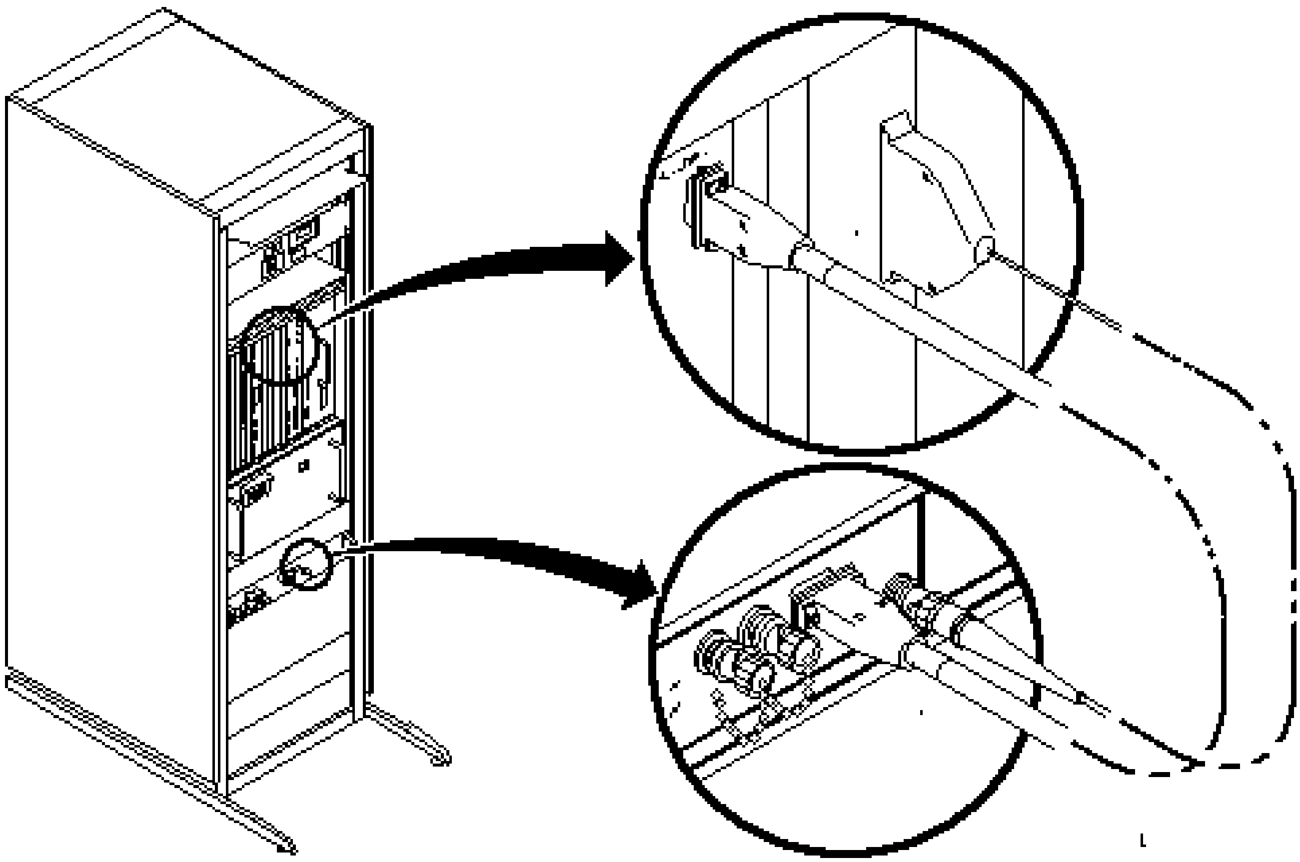
Description - This material describes the theory of operation of the EPI Fast Receiver, which, with its power supply, is located in the systems cabinet (MR2).

1- COMPONENT LOCATION AND INTERCONNECTS

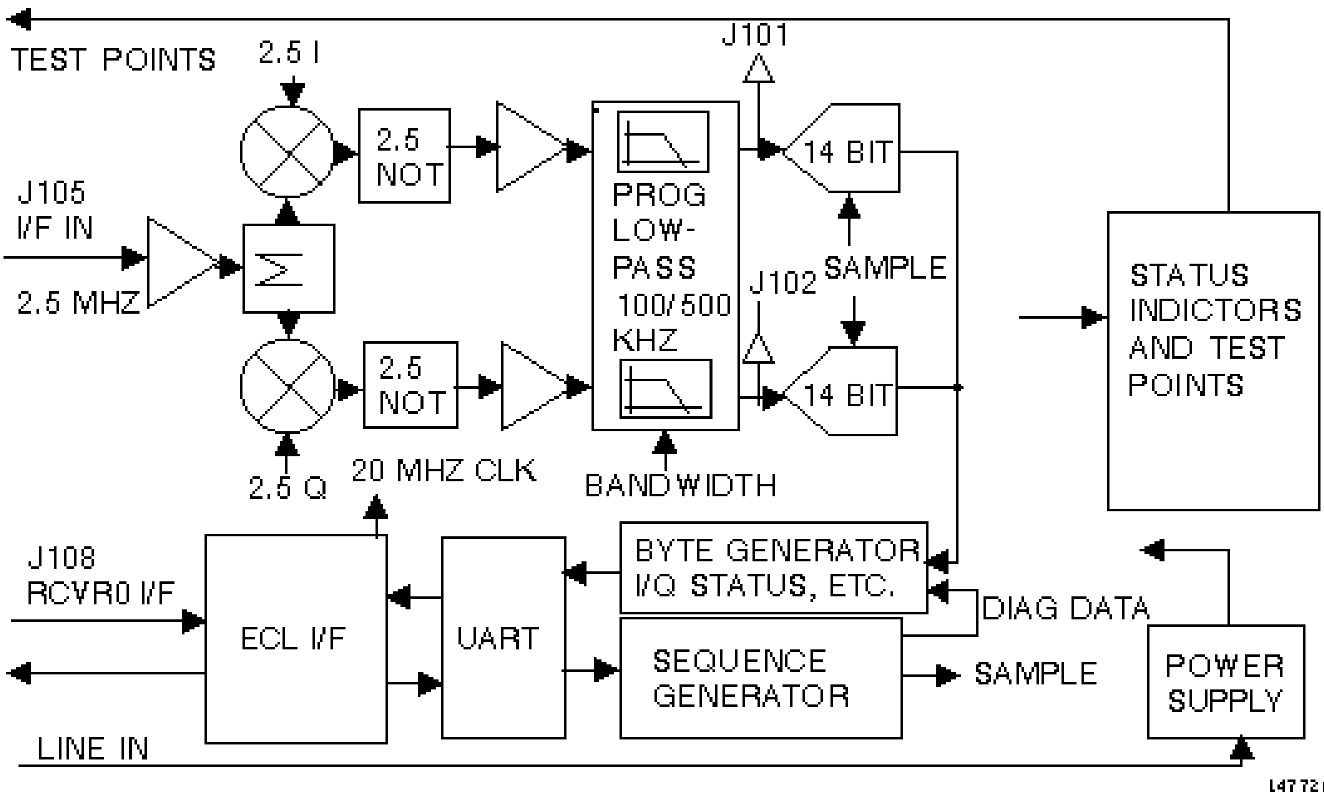
The EPI Fast Receiver is a remote receiver, and works with a modified Receiver 0 (designated RCVR_0) that is already installed in the systems cabinet. See Illustration L4771A for location and interconnects.



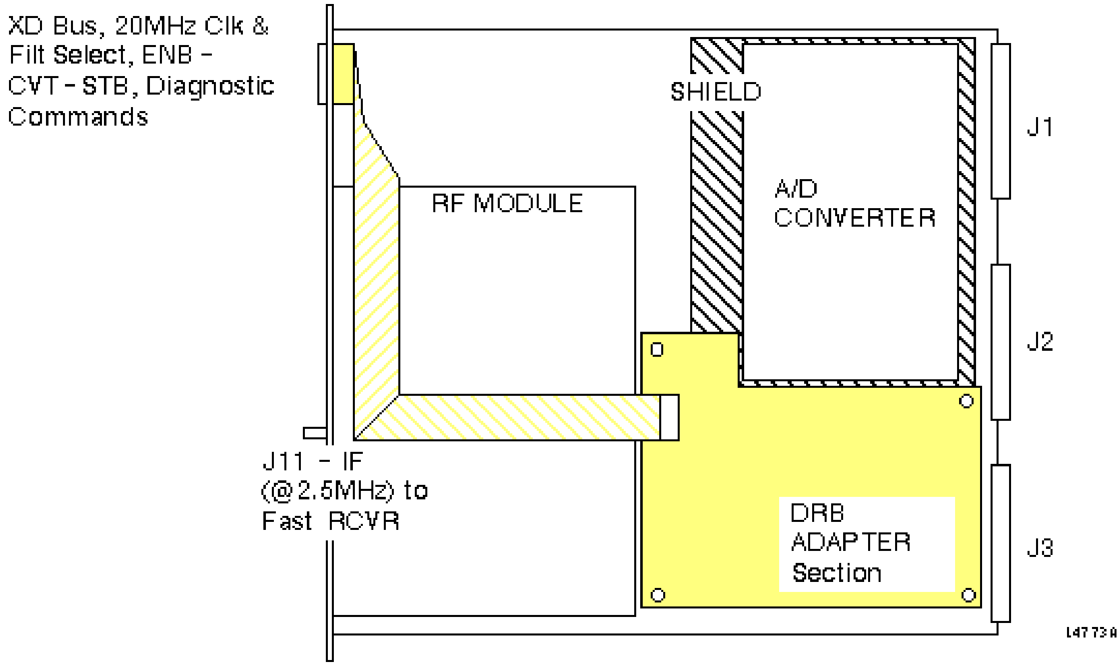
EPI FAST RECEIVER LOCATION AND INTERCONNECTS
ILLUSTRATION L4771A

2- FAST RECEIVER OVERVIEW

The Fast Receiver supports EPI (ultrafast imaging) techniques (see Illustration L4772A). The EPI fast receiver operates with a modified RCVR_0 (see Illustration L4773A). All conventional imaging techniques, including prescan operations, are performed with the standard receiver. The EPI pulse sequences optionally acquire data through the standard receiver (minimum sampling period: eight microseconds) or the EPI fast receiver (minimum sampling period: one microsecond). The Fast Receiver provides one channel of EPI reception over a ±500-kHz maximum bandwidth digitized at a 1-mHz maximum quadrature sample rate, and transfers the data to the XD bus under CERD control. The minimum quadrature sample rate is 78.4 kHz. The real-time control of the acquisition cycle is provided by the PSD via the modified RCVR_0.



FAST RECEIVER BLOCK DIAGRAM
ILLUSTRATION L4772A



RECEIVER 0 IN TPS CHASSIS
ILLUSTRATION L4773A

2-1 Limitations

The Fast Receiver is limited to one module per system (1-channel max). The Fast Receiver has an IQ pair data rate of 1.0 Mpairs/sec. The Fast Receiver provides baseband operation, analog quadrature, and five selectable twin analog, low-pass, anti-aliasing filters. The passband is selectable in steps of ±100 Hz; this coarseness is adapted for via post-processing. The Fast Receiver requires the modification of the Standard RCVR_0 (both the digital and RF portions). The Fast Receiver acquires 14-bit signed data (in contrast to the 16-bit signed data acquired by RCVR_0). Diagnostic capabilities are restricted by the limited VME and SSP interfaces on the Fast Receiver itself. The acquired composite frame size is limited to 65,536 consecutive IQ samples (1,048,576 for production release) separated in time by no less than 1 microsecond and no more than 12.7 microseconds. These consecutive IQ samples represent multiple echoes. The allowable programmable delay is no more than 12.7 microseconds.

2-2 Inputs

The Fast Receiver receives:

- Attenuated IF Signal from modified RCVR_0
- A reference clock for LO2 generation from the modified RCVR_0
- Filter settings from modified RCVR_0
- Enable, Convert, and Strobe commands from the modified RCVR_0

The Fast Receiver is powered by ±15V and ±5V analog and +5V digital.

2-3 Outputs

The Fast Receiver outputs:

- Acquired data to the XD Bus on J3 of the modified RCVR_0
- Diagnostic data to the XD Bus
- An A/D overrange bit to the XD Bus
- A data overflow bit to the XD Bus

The Fast Receiver provides an electrical output to the front panel that indicates that it is active. The data is bipolar and automatically sign-extended to 32-bit signed before transferred to the CERD over the XD bus. The CERD is set up to write these data to BAM in 16- or 32-bit signed format. The overrange indication is carried over the XD Bus to the CERD Status Register (DSR). The DSR also correlates the overrange to Receiver 4. A VME write to DSR clears the overrange condition. The overflow bit, meaning XD Bus overflow, is carried over the XD Bus to the CERD Status Register (DSR). The DSR also correlates the overflow to Receiver 4. A VME write to DSR clears the overflow condition. The Fast Receiver is equipped with LEDs, providing, at a minimum, visual indication of the following functions:

- 20 mHz Clock Data In Window, active from the beginning of the minimum delay to the beginning of the convert pulse for the last sample
- UART receive activity
- A/D data outputting to Local Module
- Diagnostic mode A
- LO present on Analog Module
- Module Sampling ongoing on Analog Module
- LP Filter Bandwidth setting on Analog Module

3- FAST RECEIVER

3-1 Operations And Control

The Fast Receiver consists of three sections: the Analog Module, Remote Digital Module, and Local RCVR_0 Adapter. The Local RCVR_0 Adapter is attached to a modified RCVR_0 board. (The RCVR_0 is modified to provide for electrical interconnection and physical mounting of the Local RCVR_0 Adapter and its cabling.) The Fast Receiver can be controlled either by a PSD executing on IPG over the SSP Bus, or by a program executed on TPS/ISE over the VME Bus. Data acquired by the Fast Receiver are transmitted to the CERD over the XD Bus, and then are transferred for storage in BAM via the VSB Bus.

3-2 Analog Module

The Fast Receiver input on the Analog Module is connected to the RCVR_0 RF Module. The signal is already attenuated by the RCVR_0 R1 analog gain. This gain factor is programmable in steps of 6 dB via the RCVR_0 R1 control register. The Analog Module performs Baseband Conversion and Quadrature Generation of the signal in the analog domain. After the Quadrature Generation, the I and Q signals respectively are passed through separate, identical programmable analog low-pass filters. These low-pass filters are programmable, with cut-off frequencies at ±100, ±200, ±300, ±400, or ±500 kHz. Next the I and Q signals are digitized using two 14-bit bipolar A/D converters running synchronously. The range of the A/D converters is [–8192,8191] inclusively. The sampling frequency of the A/Ds is programmable, with a maximum frequency of 1 mHz. The A/D converters also detect and report any data overrange in real time.

3-3 Digital Module Including Recorder

The Digital Module controls the acquisition according to commands received remotely from the Local RCVR_0 Adapter. The Digital Module functionally consists of the following key parts: The recorder, the UART and command interface, and the parallel-to-serial converter. The recorder controls and sequences an acquisition according to the PSD setup. It controls the programmable sample time, the number of points to be digitized, and the delay time between the Begin Sequencing command from the PSD and the actual first convert. Convert pulses to the Analog Module are created, and the resultant MR data are read and converted to serial format to be sent to the Local RCVR_0 adapter. Two extra bits are transmitted per data item, making each data item 16-bits. One of the extra items is the overrange bit. Convert pulses are created according to how the recorder has been set up. The recorder controls the interval between the Begin Sequencing command and the first convert pulse, as well as the sampling frequency and the number of points to acquire per frame. The Digital Module also controls the Analog Module filter selection.

3-4 Local Receiver 0

The Local RCVR_0 Adapter is connected to the RCVR_0 board. All commands written to the RCVR_0 Filter Select register that apply to the Fast Receiver are transmitted to the Digital Module by means of a UART. In order to keep up with the SSP bus, a command can be read every microsecond. The Local RCVR_0 Adapter receives MR data from the Digital Module in serial format. These data are sign-extended and are converted to parallel format for transfer over the XD bus to the CERD. The XD bus is a 16-bit data bus over which one IQ-pair is transferred in four segments: I_{lo}, I_{hi}, Q_{lo}, and Q_{hi}. The Local RCVR_0 Adapter has the ability to buffer one data sample before outputting it the XD bus. This minimizes the risk of an overflow condition. The CERD receives the MR data from the Fast Receiver as RCVR_4. In order to acquire these data properly, the CERD must be programmed with Word Count, Memory Out address, and Poll Receiver 4. Data format may be 16- or 32-bit arbitrarily set via the CERD Command Register.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 2, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.